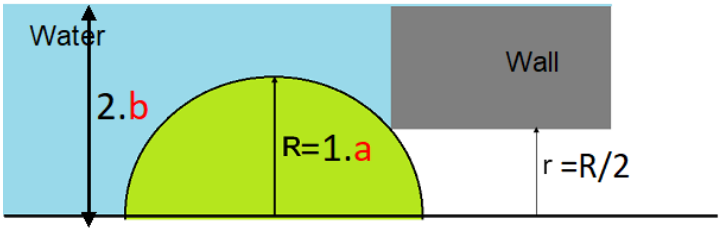
2022-2023 SPRING SEMESTER FLUID MECHANICS MIDTERM1, Date:25/04/2023

Name: **Surname:** **Student ID:**– – – – – – – – –
b a

QUESTION 2 (25p): For the half cylindrical gate shown in the figure below, determine a) the horizontal component and b) the vertical component of the hydrostatic pressure force acting on the unit length of the gate.
Show all the force components' pressure distribution by drawing on a figure.



a and b are the last two digits of your student ID.
Example: Student ID: 17042082 a=8 b=2
(a and b are the digits after the comma)